

Class Test Number Systems-1

05 June 2026 05:13

Full Marks: 20

Time: 30 mins

Instructions:

1. Answer all questions.
2. Show all conversion and arithmetic steps where applicable.
3. Calculators are not allowed.

Section A: Short Questions (1 × 5 = 5 Marks)

1. Define the following number systems:
 - Decimal
 - Binary
 - Octal
 - Hexadecimal
2. What is the base (radix) of the hexadecimal number system?
3. Convert the decimal number 25 to binary.
4. Convert the binary number 1010_2 to decimal.
5. Write the hexadecimal equivalent of decimal 15.

Section B: Conversion Problems (2 × 5 = 10 Marks)

6. Convert $(101101)_2$ to decimal. (2 Marks)
7. Convert $(57)_8$ to binary. (2 Marks)
8. Convert $(3A)_{16}$ to decimal. (2 Marks)
9. Convert $(156)_{10}$ to binary. (2 Marks)
10. Convert $(11110110)_2$ to hexadecimal. (2 Marks)

Section C: Arithmetic Operations (5 Marks)

11. Perform the following binary arithmetic operations:
 - a) $(1011)_2 + (1101)_2$ (2 Marks)
 - b) $(11010)_2 - (10101)_2$ (2 Marks)
 - c) $(101)_2 \times (11)_2$ (1 Mark)